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Electric compressor

Abstract

An electric compressor (1) includes an electric motor (5) including a stator core (51), a stator coil (52) formed by winding a conducting wire (60) around the stator core (51), and a rotor (53) attached to a rotating shaft (4), the electric motor (5) being for rotating the rotating shaft (4) via the rotor (53) through energization of the stator coil (52), a compression mechanism (3) that is driven by the rotating shaft (4) to compress a refrigerant, and a housing (2) including an intake port (21a) and a discharge port (24a), the housing (2) being for housing the rotating shaft (4), the electric motor (5), and the compression mechanism (3), where the refrigerant drawn in by the intake port (21a) is compressed by the compression mechanism (3) and discharged from the discharge port (24a). The electric compressor (1) includes a protection member (58) for covering a part (61), of the conducting wire (60), that faces the intake port (21a).

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to an electric compressor that includes a compression mechanism for compressing a refrigerant and an electric motor that are housed inside a housing.

BACKGROUND ART

(2) An example of conventional electric compressors is described in Patent Document 1. An electric compressor described in Patent Document 1 includes an electric motor for rotating a rotating shaft, a compression mechanism that is driven by the rotating shaft to compress a refrigerant, and a housing that includes an intake port and a discharge port and that houses the rotating shaft, the electric motor, and the compression mechanism, where a refrigerant that is drawn in by the intake port is compressed by the compression mechanism and is discharged from the discharge port. An example of the electric motor described above includes a stator core, a stator coil formed by a conducting wire wound around the stator core, and a rotor attached to the rotating shaft.

REFERENCE DOCUMENT LIST

Patent Document

(3) Patent Document 1: JP 2020-051328 A

SUMMARY OF THE INVENTION

Problem to be Solved by the Invention

(4) A fluid such as a refrigerant or a lubricating oil that is drawn in by an intake port includes contaminants (foreign materials) that is generated by metal pieces sliding against each other, for example. In relation to the electric compressor described above, in the case in which a part of the conducting wire described above faces the intake port, the contamination described above may collide with the conducting wire, possibly causing the conducting wire to be scratched.

(5) Accordingly, the present invention is aimed at preventing contamination included in a fluid that is drawn in from an intake port from directly colliding with a conducting wire.

Means for Solving the Problem

(6) According to an aspect of the present invention, there is provided an electric compressor. The electric compressor includes: an electric motor including a stator core, a stator coil formed by winding a conducting wire around the stator core, and a rotor attached to a rotating shaft, the electric motor being for rotating the rotating shaft via the rotor through energization of the stator coil; a compression mechanism that is driven by the rotating shaft to compress a refrigerant; a housing including an intake port and a discharge port, the housing being for housing the rotating shaft, the electric motor, and the compression mechanism, where the electric compressor is configured such that the refrigerant drawn in through the intake port is compressed by the compression mechanism and discharged from the discharge port. The electric compressor includes a protection member for covering a part, of the conducting wire, that faces the intake port.

Effects of the Invention

(7) According to the present invention, the part, of the conducting wire, that faces the intake port is covered by a protection member. Accordingly, contamination included in a fluid drawn in by the intake port is prevented by the protection member from colliding with the conducting wire. Therefore, the contamination is prevented from directly colliding with the conducting wire.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram showing an electric compressor according to a first embodiment of the present invention.
- (2) FIG. 2A is a perspective view of a stator according to the first embodiment.
- (3) FIG. 2B is a front view of the stator according to the first embodiment.
- (4) FIG. 3A is a perspective view of an insulator member according to the first embodiment.
- (5) FIG. 3B is a front view of the insulator member according to the first embodiment.
- (6) FIG. 4 is a side view of the insulator member according to the first embodiment.
- (7) FIG. 5A is a perspective view of a stator according to a second embodiment of the present invention.
- (8) FIG. 5B is a front view of the stator according to the second embodiment of the present invention.

MODE FOR CARRYING OUT THE INVENTION

- (9) Hereinafter, embodiments of the present invention will be described with reference to the appended drawings.
- (10) FIG. 1 is a diagram showing an electric compressor **1** according to a first embodiment of the present invention. The electric compressor **1** is a horizontal inverter-integrated electric compressor in which a compression mechanism **3** and an electric motor **5** are serially disposed in a horizontal direction inside a housing **2**, and in which an inverter **8** as a motor drive circuit is integrally provided. The electric compressor **1** is used in an air conditioner for a vehicle, for example, and forms a refrigeration cycle device by being integrated, together with a condenser, a pressure reducer (for example, an expansion valve), and an evaporator, into a refrigerant circulation channel for circulating a refrigerant. Additionally, in the present embodiment, a refrigerating machine oil (a lubricating oil) circulates through the refrigerant circulation channel, together with the refrigerant.
- (11) As shown in FIG. 1, the housing **2** of the electric compressor **1** includes a main housing **21**, an inverter housing **22**, an inverter cover **23**, and a discharge housing **24**. The main housing **21**, the inverter housing **22**, the inverter cover **23**, and the discharge housing **24** are formed of a metal material, for example.
- (12) The main housing **21** is formed into a cylindrical shape. An intake port **21a** for drawing the refrigerant from the evaporator side is formed on a peripheral wall of the main housing **21**, on one end side (a left side in FIG. 1). In the present embodiment, the intake port **21a** is provided at an upper part of the main housing **21**, and is open at a top. The compression mechanism **3**, a rotating shaft **4**, and the electric motor **5** are housed inside the main housing **21**.
- (13) An opening of the main housing **21** on the one end side is blocked by the inverter housing **22**. The inverter housing **22** is fastened to the main housing **21** by a bolt (not shown). The inverter housing **22** is formed into a bottomed cylindrical shape, and a bottom wall portion thereof forms a partition wall **7** that separates an inside of the main housing **21** from an inside of the inverter housing **22**.
- (14) The inverter **8** as a motor drive circuit is housed inside the inverter housing **22**. An opening of the inverter housing **22** opposite the bottom wall portion is blocked by the inverter cover **23**. The inverter cover **23** is fastened to the inverter housing **22** by a bolt (not shown).
- (15) An opening of the main housing **21** on another end side is blocked by the discharge housing **24**. The discharge housing **24** is fastened to the main housing **21** by a bolt (not shown). In the discharge housing **24**, a discharge port **24a** for discharging the refrigerant to the condenser side, and a discharge channel **24b** for guiding the refrigerant to the discharge port **24a** are formed. In the present embodiment, the discharge port **24a** is provided at an upper part of the discharge housing **24**, and is open at a top.

(16) The compression mechanism **3** is disposed inside the main housing **21**, on the discharge housing **24** side. The compression mechanism **3** is driven by rotation of the rotating shaft **4**, and takes in and compresses the refrigerant that is drawn into the main housing **21** by the intake port **21a** and discharges the compressed refrigerant. The refrigerant that is discharged from the compression mechanism **3** is guided to the discharge port **24a** by the discharge channel **24b**, and is discharged from the discharge port **24a**. Although it is not so limited, a scroll compression mechanism including a fixed scroll and a movable scroll may be adopted as the compression mechanism **3**, for example.

(17) The rotating shaft **4** extends in an axial direction of the main housing **21**. The rotating shaft **4** is rotatably supported by a bearing, not shown, inside the main housing **21** in a state in which one end of the rotating shaft **4** is coupled to the compression mechanism **3** by a coupling portion **9**.

(18) Additionally, in the case in which the compression mechanism **3** is the scroll compression mechanism, the coupling portion **9** may be a crank mechanism for converting a rotating motion of the rotating shaft **4** into an orbiting motion of the movable scroll. The scroll compression mechanism as the compression mechanism **3** takes in and compresses the refrigerant and discharges the compressed refrigerant by the orbiting motion of the movable scroll relative to the fixed scroll caused by rotation of the rotating shaft **4**.

(19) The electric motor **5** is disposed inside the main housing **21**, on the inverter housing **22** side. That is, the electric motor **5** is disposed inside the main housing **21**, at a position closer to the intake port **21a** than the compression mechanism **3**. The electric motor **5** causes the rotating shaft **4** to rotate. In other words, the electric motor **5** drives the compression mechanism **3** via the rotating shaft **4**.

(20) FIGS. **2A** and **2B** are a perspective view and a front view of a stator **50** constituting the electric motor **5**. As shown in FIGS. **1**, **2A**, and **2B**, the electric motor **5** includes the stator **50** including a stator core **51** and stator coils **52**, and a rotor **53**.

(21) The stator core **51** is a magnetic body, and is formed into a cylindrical shape. The stator core **51** is supported by an inner wall of the main housing **21**. The stator core **51** includes a plurality of (in this case, 12) teeth **511** that protrude radially inward (toward the rotating shaft **4**). The plurality of teeth **511** are disposed at regular intervals in a circumferential direction.

(22) The stator coils **52** are formed by concentrated winding of the conducting wire **60** around the stator core **51**. More specifically, the stator coils **52** are formed by the conducting wire **60** being wound around each of the plurality of teeth **511** of the stator core **51**. Additionally, for example, the conducting wire **60** is a linear conductor (a core wire) on which insulation coating is formed. In other words, the conducting wire **60** is a winding of the stator **50**.

(23) The stator **50** further includes an insulator **54** for electrically insulating the stator core **51** and the stator coil **52**. The insulator **54** is formed of a resin material having an insulating property. In the present embodiment, the insulator **54** is divided into two insulator members **541**, **542**.

(24) FIGS. **3A** and **3B** are a perspective view and a front view of the insulator member **541**. FIG. **4** is a side view of the insulator member **541**.

(25) The insulator member **541** includes an annular base portion **541a**, and protruding portion **541b** that extends radially inward from the base portion **541a**. The protruding portion **541b** is interposed between the tooth **511** and the stator coil **52** to insulate the two. A configuration of the insulator member **542** may be the same as that of the insulator member **541**.

(26) On an end portion, of the base portion **541a** of the insulator member **541**, on a side of a coil end of the stator coil **52**, a plurality of (in this case, 12) projecting portions **55** are formed at regular intervals in the circumferential direction to hook the conducting wire **60**.

(27) A part **P** shown in FIGS. **1** to **4** faces the intake port **21a**.

(28) The rotor **53** is disposed on a radially inward side of the stator core **51** (the plurality of teeth **511**). A permanent magnet, not shown, is embedded in the rotor **53**. The rotor **53** is formed into a cylindrical shape, and is fixed to the rotating shaft **4** in a state in which the rotating shaft **4** is

inserted in a hollow part of the rotor **53**. That is, the rotor **53** is attached to the rotating shaft **4** to rotate together with the rotating shaft **4**.

(29) The inverter **8** includes various electronic components such as a smoothing capacitor and a power module including a plurality of power switching elements, and a circuit board at which the various electronic components are mounted. The inverter **8** is connected to an external power source (such as a vehicle-mounted battery) via a first power line, not shown, and is connected to the electric motor **5** (the stator coil **52**) via a second power line, not shown, that penetrates through the partition wall **7** in an airtight and liquid-tight manner.

(30) When a power supply voltage is applied to the inverter **8** by the external power source, the inverter **8** outputs an alternating current to the stator coils **52** of the electric motor **5**. That is, the stator coils **52** are energized. When the stator coils **52** are energized, a rotating magnetic field is generated, and the rotor **53** rotates in synchronization with the rotating magnetic field that is generated. The rotating shaft **4** is thereby rotated to drive the compression mechanism **3**. Furthermore, the refrigerant is drawn into the main housing **21** by the intake port **21a**.

(31) The refrigerant that is drawn into the main housing **21** by the intake port **21a** first passes through the electric motor **5**, and then through the compression mechanism **3**. That is, the refrigerant that is drawn in by the intake port **21a** sequentially passes through the electric motor **5** and the compression mechanism **3**. The refrigerant that is drawn in by the intake port **21a** cools the electric motor **5** at the time of passing through the electric motor **5**, and is compressed by the compression mechanism **3** at the time of passing through the compression mechanism **3**, and the compressed refrigerant is discharged from the discharge port **24a**.

(32) In the present embodiment, a protection member **58** that covers a part **61**, of the conducting wire **60**, that faces the intake port **21a** is provided at the base portion **541a** of the insulator member **541**. In particular, in the present embodiment, the protection member **58** is formed integrally with the base portion **541a** of the insulator member **541**. Additionally, in the present embodiment, the part **61** of the conducting wire **60** is also a part that connects the stator coils **52** that are adjacent to each other in the circumferential direction.

(33) The protection member **58** has a U-shaped or C-shaped wall shape when seen from a front of the insulator member **541**. The protection member **58**, for example, is formed from a pair of wall portions **58a**, **58b** that face each other with a space in between, and a wall portion **58c** that extends between the wall portions **58a**, **58b**. The wall portions **58a**, **58b** each extend radially outward from the base portion **541a** of the insulator member **541**. The wall portion **58c** extends substantially in parallel with the base portion **541a** of the insulator member **541**.

(34) The wall portion **58c** of the protection member **58** faces the intake port **21a**. When seen by the intake port **21a**, the part **61** of the conducting wire **60** is hidden by the wall portion **58c** of the protection member **58**. That is, the part **61** of the conducting wire **60** is disposed between the projecting portion **55** of the insulator member **541** and the wall portion **58c** of the protection member **58**. Furthermore, particularly in the present embodiment, the protection member **58** covers the part **61** of the conducting wire **60** and the projecting portion **55**.

(35) Additionally, in the present embodiment, the shape of the protection member **58** is a U-shaped or C-shaped wall shape mentioned above, but the shape of the protection member **58** is not limited thereto. For example, the shape of the protection member **58** may be an arc-shaped wall shape when seen from the front of the insulator member **541**.

(36) According to the present embodiment, the electric compressor **1** includes the electric motor **5** including the stator core **51**, the stator coil **52** formed by winding the conducting wire **60** around the stator core **51**, and the rotor **53** attached to the rotating shaft **4**, the electric motor **5** being for rotating the rotating shaft **4** via the rotor **53** through energization of the stator coil **52**; the compression mechanism **3** that is driven by the rotating shaft **4** to compress a refrigerant; and the housing **2** including the intake port **21a** and the discharge port **24a**, the housing **2** being for housing the rotating shaft **4**, the electric motor **5**, and the compression mechanism **3**. The electric

compressor **1** is configured such that the refrigerant drawn in by the intake port **21a** is compressed by the compression mechanism **3** and is discharged from the discharge port **24a**. The electric motor **5** is an inner rotor motor at which the rotor **53** is disposed on a radially inner side of the stator core **51** having a cylindrical shape. The electric compressor **1** includes the protection member **58** for covering the part **61**, of the conducting wire **60**, that faces the intake port **21a**. Accordingly, contaminants included in a fluid drawn in by the intake port **21a** is prevented, by the protection member **58**, from colliding with the conducting wire **60**. Accordingly, the contaminants are prevented from directly colliding with the conducting wire **60**.

(37) Furthermore, according to the present embodiment, the electric motor **5** further includes the insulator **54** for insulating the stator core **51** and the stator coil **52**. The protection member **58** is provided at the insulator **54**. The protection member **58** is preferably integrally formed with the insulator **54**. This allows the protection member **58** to be assembled together with the insulator **54**.

(38) Furthermore, according to the present embodiment, the part **61** of the conducting wire **60** is disposed between the projecting portion **55** of the insulator **54** and the protection member **58**. Accordingly, the part **61** of the conducting wire **60** may be well protected from the contamination included in the fluid drawn in by the intake port **21a**.

(39) Furthermore, according to the present embodiment, the protection member **58** has a wall shape. Accordingly, the protection member **58** may be achieved by a simple structure.

(40) Next, a second embodiment of the present invention will be described with reference to FIGS. 5A and 5B.

(41) FIGS. 5A and 5B are a perspective view and a front view of the stator **50** according to the present embodiment. Differences from the first embodiment described above will be described.

(42) In the present embodiment, a protection member **59** is provided instead of the protection member **58** described above. The protection member **59** is formed by a molding resin. In other words, the part **61** of the conducting wire **60** is covered by molding (sealing) of the resin, and the molded part functions as the protection member **59**. The resin (molding resin) includes a thermosetting resin having an insulating property, and includes an epoxy resin, for example.

(43) The protection member **59** faces the intake port **21a**. When seen from the intake port **21a**, the part **61** of the conducting wire **60** is hidden by the protection member **59**. The part **61** of the conducting wire **60** may be disposed between the projecting portion **55** of the insulator member **541** and the protection member **59**. In particular, in the present embodiment, the protection member **59** may cover the part **61** of the conducting wire **60** and the projecting portion **55**.

(44) In particular, according to the present embodiment, the protection member **59** is formed by a molding resin that covers the part **61** of the conducting wire **60**. Accordingly, the part **61** of the conducting wire **60** may be easily covered by molding the resin, and thus, the part **61** of the conducting wire **60** may be easily protected.

(45) Additionally, in the first and second embodiments, the electric compressor according to the present invention is applied to a horizontal electric compressor at which the compression mechanism and the electric motor are serially disposed in the horizontal direction inside the housing, but it is apparent that the electric compressor according to the present invention may also be applied to a vertical electric compressor at which the compression mechanism and the electric motor are serially disposed in a vertical direction inside the housing.

(46) Heretofore, the embodiments of the present invention have been described, but the present invention is not limited to the embodiments described above, and modifications and changes can, of course, be made based on the technical idea of the present invention.

REFERENCE SYMBOL LIST

(47) **1** Electric compressor **2** Housing **3** Compression mechanism **4** Rotating shaft **5** Electric motor **8** Inverter **21a** Intake port **24a** Discharge port **50** Stator **51** Stator core **52** Stator coil **53** Rotor **54** Insulator **55** Projecting portion **58** Protection member **58a**, **58b**, **58c** Wall portion **59** Protection

Claims

1. An electric compressor comprising: an electric motor including a stator core, a stator coil formed by winding a conducting wire around the stator core, a rotor attached to a rotating shaft, and an insulator for insulating the stator core and the stator coil, the electric motor being for rotating the rotating shaft via the rotor through energization of the stator coil; a compression mechanism that is driven by the rotating shaft to compress a refrigerant; a housing including an intake port and a discharge port, the housing being for housing the rotating shaft, the electric motor, and the compression mechanism, wherein the electric compressor is configured such that the refrigerant drawn in by the intake port is compressed by the compression mechanism and is discharged from the discharge port, and a protection member for covering a part, of the conducting wire, that faces the intake port is provided, the protection member is provided on the insulator, the insulator has a projecting portion to hook the conducting wire, and the part is disposed between the projecting portion and the protection member.
 2. The electric compressor according to claim 1, wherein the electric motor is an inner rotor motor at which the rotor is disposed on a radially inner side of the stator core having a cylindrical shape.
 3. The electric compressor according to claim 1, wherein the protection member is formed by a molding resin that covers the part.
 4. The electric compressor according to claim 1, wherein the protection member is integrally formed with the insulator.
 5. The electric compressor according to claim 1, wherein the protection member has a wall shape.
 6. The electric compressor according to claim 1, wherein a void is formed between the part and the protection member.
 7. The electric compressor according to claim 1, wherein the insulator is divided into a plurality of insulator members, and wherein the protection member is provided on one of the plurality of insulator members.
 8. The electric compressor according to claim 1, wherein contaminants included in the refrigerant drawn in by the intake port are prevented, by the protection member, from colliding with the conducting wire.
 9. The electric compressor according to claim 1, wherein the insulator further includes an annular base portion defining a center axis direction, and the projecting portion extends along the center axis direction from the annular base portion.
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